LECTURE 1

From Data to Decisions: Statistical Models, Loss, and Risk

STAT S4204 | Statistical Inference

Monday, 6 July 2026

Far better an approximate answer to the right question, which is often vague, than an exact answer to the wrong question, which can always be made precise.

John W. Tukey, "The Future of Data Analysis" (1962)

Contents of this lecture

Overview. The course opens where applied statistics leaves off: a study has a goal, data, and software output—and none of those, by themselves, say anything about reality. This lecture builds the bridge. After a review of random variables, distributions, and realizations (and, later, of expectation up through iterated expectations), the data's unknown joint distribution is organized into a *statistical model*, the study's purpose becomes an *analytic goal* expressed in the model's parameters, and the choice of method becomes a decision problem: an action space, a loss function, and—as the figure of merit—the *risk function*, the expected loss as a function of the parameter. Admissibility and the Bayes principle give the two classical answers to what "good" means, and both are put to work on a two-arm clinical trial in which two estimators of a cure-rate difference compete. The companion text is Hoel–Port–Stone, Chapter 1 (HPS 1).

1.1 Probability Review I: Randomness, Random Variables, Distributions, and Realizations

Concept 1.1.1

The point of statistical theory

"The point of statistical theory is to inform the application of statistics." The lecture therefore begins with applied statistics—what a data analysis is *for*—and works backward to the theory that disciplines it.

Remark 1.1.2

Randomness—does it exist?

The review opens with a philosophical aside: whether randomness "exists" in the world or only describes our ignorance is left unresolved—and is beside the point. Probability is the *model* we adopt for quantities whose values we cannot predict; everything in this course is a statement within that model.

Definition 1.1.3

Sample space

The *sample space* S is the set of all possible outcomes of the random phenomenon being modeled. Events are subsets of S, and probabilities are assigned to events in accordance with the axioms of probability.

Definition 1.1.4

Random variable

In the real world, a *random variable* is a numerical quantity whose value is determined by the outcome of a random phenomenon—a measurement not yet made. Mathematically, a random variable is a (measurable) function

$$X : S \to \mathbb{R},$$

assigning a number $X(s)$ to each outcome $s \in S$. The data of a study are random variables; so is anything computed from them.

Definition 1.1.5

Distribution

The *distribution* of a random variable X is the assignment

$$A \longmapsto P(X \in A)$$

of a probability to each (measurable) set $A \subseteq \mathbb{R}$: the complete description, in advance of observation, of how likely X is to fall anywhere. It is determined by the cumulative distribution function $F(x) = P(X \leq x)$, and, in the discrete and continuous cases, by a probability mass function or a density.

Definition 1.1.6

Joint distribution, joint pmf, joint density

The *joint distribution* of random variables $X_1, \ldots, X_n$ assigns a probability $P\big((X_1, \ldots, X_n) \in B\big)$ to each (measurable) $B \subseteq \mathbb{R}^n$; it carries every dependence among the variables, and it determines—but is not in general determined by—the marginal distribution of each X_i. In the discrete case it is given by the *joint probability mass function* $p(x_1, \ldots, x_n) = P(X_1 = x_1, \ldots, X_n = x_n)$; in the continuous case by a *joint density*[1] $f \geq 0$ with

$$P\big((X_1, \ldots, X_n) \in B\big) \;=\; \int_B f(x_1, \ldots, x_n)\, dx_1 \cdots dx_n.$$

Definition 1.1.7

Realization

If the outcome of the experiment is $s \in S$, the *realization* of the random variable X is the number $x = X(s)$: the value actually observed. Random variables are written with capital letters and their realizations in lower case. Your data set is a realization of the random variables the model describes.

Example 1.1.8

The same coin flipped twice

Flip the same coin twice, and let X_1 and X_2 be the head-indicators of the two flips. The deck poses three questions. *Are the random variables the same?* No: X_1 and X_2 are different functions on the sample space $S = \{HH, HT, TH, TT\}$—the first reads off the first coordinate, the second the second. *Are the distributions the same?* Yes: each is Bernoulli with the same head-probability, because it is the same coin. *Are the realizations the same?* Not necessarily: the observed pair may be $(1, 0)$. Distinct random variables can share one distribution and still realize differently—the three notions of 1.1.4, 1.1.5, and 1.1.7 are genuinely different.

1.2 The Applied-Statistics Pipeline

Concept 1.2.1

The goal of the study versus the data

"If you are doing a statistical analysis, there should be something you are trying to learn… NOT something about your data, something about what your data says about reality." The goal of the study and the data are separate objects; no arrow connects them directly.

Concept 1.2.2

Methods and output

[1] What a density *is*: not a probability but a probability *rate*—probability per unit length (per unit area or volume, for joint densities). Probabilities come only from integrating it over regions, so for small dx, $f(x)\,dx \approx P(x \leq X \leq x + dx)$. Two consequences that trip people up: a density can exceed 1 (the uniform distribution on $[0, \frac{1}{2}]$ has $f \equiv 2$ there), and every single point carries probability zero ($\int_x^x f = 0$), so for a continuous random variable $f(x)$ is emphatically *not* $P(X = x)$. The pmf, by contrast, *is* a probability at each point; the twin formulas—sums against p, integrals against f—conceal that difference in kind.

Statistical methods are applied to the data—through a package (R, SAS, SPSS, Stata, ...), through modules (SciPy, NumPy, Pandas, ...), "or ... AI?"—and "the application of methods produces output..."

Concept 1.2.3

Interpretation

"These results do not directly speak to the goal!" They must be *interpreted*, and it is the interpretation of the results *in light of where the data came from*—causality and representativeness—that allows conclusions about reality.

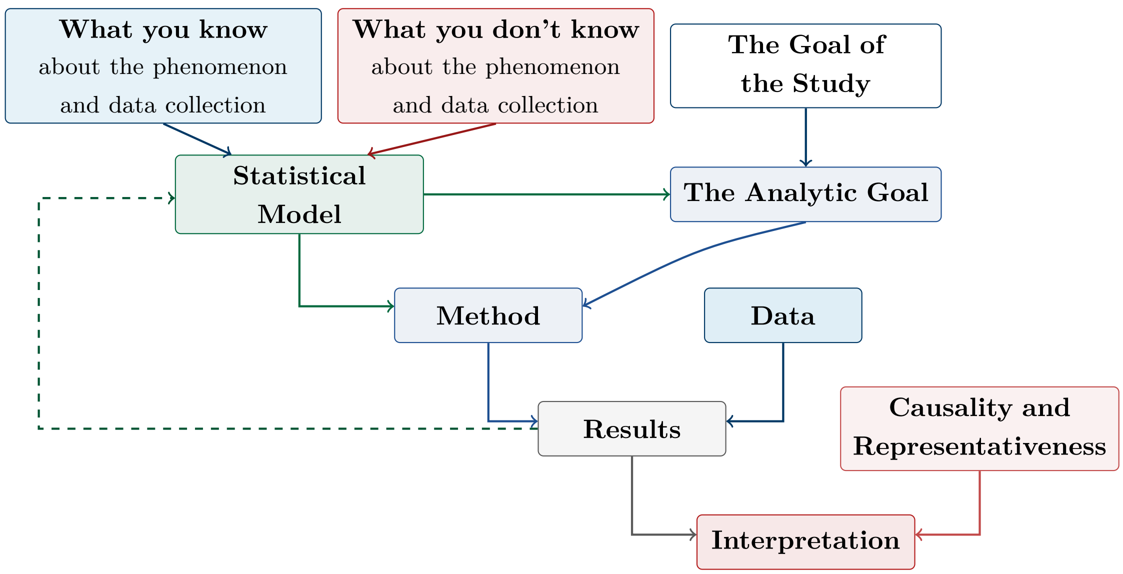

Figure 1.2.4. THE APPLIED-STATISTICS PIPELINE. What you know and what you don't know about the phenomenon and the data collection feed the statistical model; the model and the goal of the study produce the analytic goal; model and analytic goal select the method; the method applied to the data produces results, which—interpreted in light of causality and representativeness—yield conclusions. The dashed arrow is the feedback from results back into the model. Recreated from the lecture deck (1.2.1–1.2.3; cf. 1.3.1, 1.3.5).

1.3 The Statistical Model and the Analytic Goal

Definition 1.3.1

The statistical model

If your data are random, they have a joint distribution (1.1.6). What can you learn from your data? "The data! AND THE DISTRIBUTION OF THE DATA?! And that is all!" If the study has a goal, there must be something you do not know about that distribution—so more than one distribution must be possible. *The statistical model is the set of all possible distributions of the data.* HPS 1

Definition 1.3.2

Parameterization

It is common to *index* the family of distributions: a *parameterization* assigns to each value θ of a *parameter space* Θ a distribution in the model, so the model may be written

$$\{\, p^{\theta} \text{ or } f^{\theta} \;:\; \theta \in \Theta \,\}$$

with p^{θ} a pmf or f^{θ} a density for the data.

Definition 1.3.3

The analytic goal

The *analytic goal* is an expression of the purpose of the study in terms of the parameters of the model. The scientific question, which lives in the real world, is translated into a statement about θ.

Example 1.3.4

A two-arm clinical trial (the questions)

A clinical trial compares a treatment arm with a placebo arm, measuring cure or failure for each patient; the goal is to determine whether the treatment is efficacious. What is the statistical model? What is the analytic goal? The formal answers are assembled in 1.8.4, once decision theory has supplied the vocabulary.

Concept 1.3.5

Where does the statistical model come from?

From *what you know* about the phenomenon and the data collection (which distributions are plausible) and *what you don't know* (which is why the model contains more than one distribution)—the two inputs at the top of Figure 1.2.4. The dashed feedback arrow records that results can send you back to revise the model.

Remark 1.3.6

What this class is about

Mostly: how do we get from *a model and an analytic goal* to *a good method*—and first, what do we mean by "good"? In the deck's own strongest emphasis, these are "an answer, of sorts, to the question: what is it all about, extracting information from data?"

1.4 Probability Review II: Expectation, Conditional Expectation, and Iterated Expectations

The roadmap for the rest of the lecture: "First, all the theory. Then a list of the essential points. Then an illustration (the clinical trial)." But first, some more probability review.

Definition 1.4.1

Expectation

Expectations belong to *distributions*: a random variable has an expectation through its distribution. For discrete Y with pmf p and for continuous Y with density f,

$$\mathrm{E}\{Y\} \;=\; \sum_y y\,p(y) \qquad \text{or} \qquad \mathrm{E}\{Y\} \;=\; \int y\,f(y)\,dy,$$

provided the sum or integral converges absolutely[2]: the probability-weighted average of the values of Y.[3]

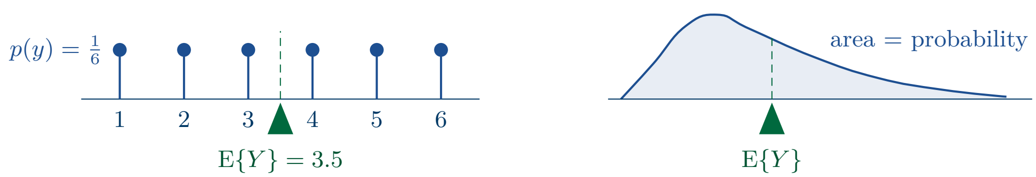

Figure 1.4.2. THE EXPECTATION AS A BALANCING POINT. Left: the fair die puts mass $\frac{1}{6}$ at each of $1, \ldots, 6$; the beam balances at $\mathrm{E}\{Y\} = 3.5$, a value the die never shows. Right: for a skewed density, probability is area, and the long right tail carries enough of it to drag the balance point past the peak. In both cases $\mathrm{E}\{Y\}$ is the center of mass of the distribution—the unique c with $\mathrm{E}\{Y - c\} = 0$ (1.4.1).

Concept 1.4.3

The problem the next theorem solves

You know the distribution of Y, and you need the expectation not of Y itself but of some transformed quantity $g(Y)$: Y^2 on the way to a variance, a loss $L(\delta(Y), \theta)$ on the way to a risk, an estimator $(1 + X)/(2 + n)$ built from a count. Now $g(Y)$ is a random variable in its own right, and 1.4.1 is unambiguous about what its expectation means: weight each value z that $g(Y)$ can take by the probability that it takes it—compute against the distribution *of* $g(Y)$. Taken literally, that instruction opens a preliminary project: derive that distribution. List the values g can produce; for each one, find all the outcomes y that g sends there and merge their probabilities; only then average. The project is real work, and it would have to be redone for every new g. The next theorem says the project is unnecessary: the same number can be read directly off the distribution you already hold, by weighting

[2] *Absolute* convergence means the sum or integral of $|y|$ is finite—$\sum_y |y|\,p(y) < \infty$ or $\int |y|\,f(y)\,dy < \infty$, i.e. $\mathrm{E}\{|Y|\} < \infty$—not merely that the signed sum happens to settle. The assumption earns its keep twice. First, a distribution's values come in no privileged order, and by Riemann's rearrangement theorem a conditionally convergent series can be reordered to reach *any* value—so without absolute convergence, "the" expectation would depend on bookkeeping. Take Y with $P\big(Y = (-1)^{k+1}2^k/k\big) = 2^{-k}$ for $k = 1, 2, \ldots$: summed in the natural order, $\sum_k (-1)^{k+1}/k = \ln 2$, yet $\mathrm{E}\{|Y|\} = \sum_k 1/k = \infty$, and rearranging the terms changes the answer—so such a Y is assigned *no* expectation. Second, absolute convergence is precisely the license for the regrouping and exchange-of-order steps in the proofs of 1.4.4 and 1.4.16 (for integrals, via Fubini's theorem).

[3] Four conflated words, disentangled. *Expectation* is the number defined here—a property of the distribution, fixed before any data are seen. *Mean* is its synonym when speaking of a distribution (a "population mean"), but a *sample mean* is something else: it is an *average*—the arithmetic operation $\frac{1}{n}\sum_i y_i$ on a finite list of numbers, defined with no probability in sight. Averaging realized data produces a random quantity that *estimates* the expectation; only when every possible value is equally likely does the expectation reduce to the plain average of those values. Finally, "*balancing point*" is the physical reading of the same number: place the probability as mass along the line, and $\mathrm{E}\{Y\}$ is its center of mass—the unique c with $\mathrm{E}\{Y - c\} = 0$. It need not be a possible value: a fair die balances at 3.5.

$g(y)$, rather than y, by $p(y)$. The name is the profession's joke at its own expense—the formula is what everyone writes down instinctively, without noticing that, by the definition, there was something to prove; the theorem is why the instinct is correct. And it earns its keep at once: the risk function of 1.5.9 *is* this formula with g the loss, a variance is this formula with $g(y) = (y-\mu)^2$, and its bivariate form, $\mathrm{E}\{g(X,Y)\} = \sum_x \sum_y g(x,y)\,p(x,y)$, is behind every covariance.

Theorem 1.4.4

The law of the unconscious statistician

Let Y be discrete with pmf p, and let g be a real function. Then

$$\mathrm{E}\{g(Y)\} \;=\; \sum_y g(y)\,p(y)$$

(and $\mathrm{E}\{g(Y)\} = \int g(y)f(y)\,dy$ in the continuous case): the expectation of $g(Y)$ may be computed against the distribution of Y, without first finding the distribution of $g(Y)$. Equivalently: grouping the average by outcome (each y weighted by $p(y)$) and grouping it by value (each z weighted by $P(g(Y)=z)$) must agree. The theorem is a regrouping identity, and its proof is exactly that regrouping.

Proof 1.4.4

Proof by Regrouping of the Unconscious-Statistician Formula (discrete case).

(1) Let $Z = g(Y)$, a random variable with pmf $p_Z(z) = P(g(Y)=z) = \sum_{y\colon g(y)=z} p(y)$.

(2) By 1.4.1,
$$\mathrm{E}\{Z\} \;=\; \sum_z z\,p_Z(z) \;=\; \sum_z z \sum_{y\colon g(y)=z} p(y).$$

(3) Each outcome y appears in exactly one inner sum, contributing $g(y)\,p(y)$; regrouping the double sum by y (legitimate by absolute convergence) gives $\mathrm{E}\{g(Y)\} = \sum_y g(y)\,p(y)$.

■

Example 1.4.5

LOTUS on a many-to-one map

Let Y be uniform on $\{-1,0,1\}$ and $g(y) = y^2$. The definition-first route derives the distribution of $Z = g(Y)$: the two outcomes -1 and 1 both land on the value 1, so their probabilities merge, $P(Z=1) = \frac{2}{3}$, while $P(Z=0) = \frac{1}{3}$; then $\mathrm{E}\{Z\} = 0\cdot\frac{1}{3} + 1\cdot\frac{2}{3} = \frac{2}{3}$. The theorem's route never mentions Z's distribution:

$$\mathrm{E}\{g(Y)\} \;=\; (-1)^2\cdot\tfrac{1}{3} + 0^2\cdot\tfrac{1}{3} + 1^2\cdot\tfrac{1}{3} \;=\; \tfrac{2}{3}.$$

Same number, different bookkeeping—and the difference is the whole content of the theorem: the first route's single term $1\cdot\frac{2}{3}$ is the second route's two terms $(-1)^2\cdot\frac{1}{3} + 1^2\cdot\frac{1}{3}$, regrouped. That merging of outcomes that share a value is all the proof of 1.4.4 does in general.

Definition 1.4.6

The joint distribution of a pair

For a pair of random variables (X, Y), the joint distribution of 1.1.6 specializes to a joint pmf $p(x, y) = P(X = x,\, Y = y)$ in the discrete case, or a joint density $f(x, y)$ in the continuous case.

Remark 1.4.7

Reading the joint: the two-way table

The joint is the complete description of the pair: it carries not only how each variable behaves alone but every trace of the dependence between them. For discrete variables it is usefully pictured as a two-way table—one row per value x, one column per value y, the cell holding $p(x, y)$—with all the cells summing to 1. The next two definitions are the table's row totals and its rescaled rows.

Definition 1.4.8

Marginal distribution

The *marginal* pmf of X recovers the distribution of X alone by summing the joint over everything that can happen to Y:

$$p_X(x) \;=\; \sum_y p(x, y)$$

(and $f_X(x) = \int f(x, y)\, dy$ in the continuous case).

Remark 1.4.9

Why "marginal," and what marginalizing forgets

In the two-way table the marginal's values are the row totals, classically written in the table's *margin*—hence the name. And marginalizing is a one-way street: the joint determines both marginals, but the marginals do not determine the joint, because everything about the dependence has been summed away (1.1.6).

Definition 1.4.10

Conditional distribution

For any x with $p_X(x) > 0$, the *conditional* pmf of Y given $X = x$ is

$$p_{Y|X}(y \mid x) \;=\; \frac{p(x, y)}{p_X(x)}$$

(and $f_{Y|X}(y \mid x) = f(x, y)/f_X(x)$ for densities, when $f_X(x) > 0$).

Remark 1.4.11

Conditioning as reweighting a slice

The recipe behind the formula: fix $X = x$, take that row of the table—the x-slice of the joint—and rescale it by its total so it sums to 1. The result is the distribution of Y for someone who has learned that $X = x$: conditioning is how a distribution absorbs information. And the formula is an old friend in new clothes: since $\{Y = y\} \cap \{X = x\}$ has probability $p(x, y)$, the display is the elementary $P(A \mid B) = P(A \cap B)/P(B)$ with $B = \{X = x\}$—applied to every event $\{Y = y\}$ at once, which is what upgrades a single conditional probability into an entire conditional distribution.

Example 1.4.12

A worked table: joint to marginal to conditional

Let $X \in \{0,1\}$ and $Y \in \{1,2,3\}$ have the joint pmf

$p(x,y)$	$y=1$	$y=2$	$y=3$	$p_X(x)$
$x=0$	0.20	0.20	0.10	0.50
$x=1$	0.10	0.15	0.25	0.50
$p_Y(y)$	0.30	0.35	0.35	1

The margins are computed, not given: the $x=1$ row sums to the marginal $p_X(1) = 0.10+0.15+0.25 = 0.50$ (1.4.8), and the column sums give $p_Y = (0.30, 0.35, 0.35)$. Conditioning on $X = 1$ rescales that row by its total (1.4.10):

$$p_{Y|X}(\,\cdot \mid 1) \;=\; \frac{(0.10,\ 0.15,\ 0.25)}{0.50} \;=\; (0.20,\ 0.30,\ 0.50),$$

which sums to 1, as a distribution must. Two things to read off. First, the conditional differs from the marginal p_Y—learning $X = 1$ tilts Y upward—so X carries information about Y: the variables are dependent. Second, the numbers quietly preview 1.4.16: here $\mathrm{E}\{Y \mid X = 0\} = 1.8$ and $\mathrm{E}\{Y \mid X = 1\} = 2.3$, and their p_X-weighted average $0.5(1.8) + 0.5(2.3) = 2.05$ is exactly $\mathrm{E}\{Y\} = 1(0.30) + 2(0.35) + 3(0.35) = 2.05$.

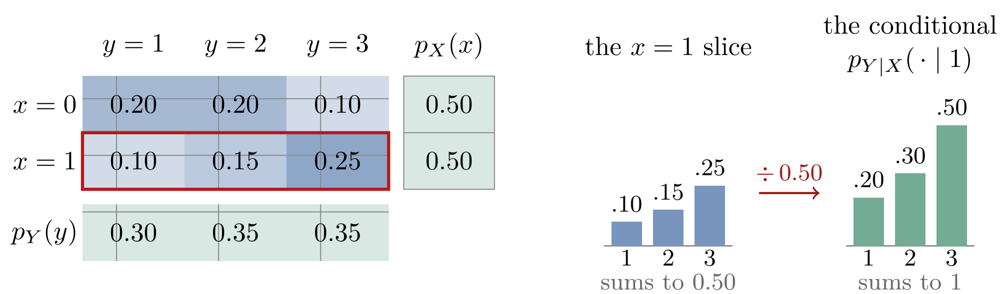

Figure 1.4.13. Slicing a joint distribution into a conditional. The worked example's joint pmf as a tinted grid (darker cells carry more probability), with its marginals in the margins; the outlined $x = 1$ row is pulled out and rescaled by its total, $p_X(1) = 0.50$, into the conditional distribution, which sums to 1. The slice-and-renormalize recipe of 1.4.11 in one picture (1.4.6, 1.4.8, 1.4.10).

Remark 1.4.14

Conditional distributions are probability distributions!

For each fixed x, the numbers $p_{Y|X}(y \mid x)$ are nonnegative and sum to 1: a conditional distribution is a genuine probability distribution, so everything defined for distributions—expectations included—applies to it verbatim. That single observation powers the next two objects.

Definition 1.4.15

Conditional expectation

The *conditional expectation* of Y given $X = x$ is the expectation of the conditional distribution (1.4.10),

$$\mathrm{E}\{Y \mid X = x\} \;=\; \sum_y y\, p_{Y|X}(y \mid x),$$

and $\mathrm{E}\{Y \mid X\}$ denotes the random variable obtained by evaluating this function of x at X.

Theorem 1.4.16

Iterated expectations

Whenever the expectations exist,

$$\mathrm{E}\{\mathrm{E}\{Y \mid X\}\} \;=\; \mathrm{E}\{Y\}.$$

Proof 1.4.16.a

Proof by Conditioning of the Iterated-Expectations Identity (discrete case).

(1) $\mathrm{E}\{Y \mid X\}$ is the function $x \mapsto \mathrm{E}\{Y \mid X = x\}$ evaluated at X, so by the law of the unconscious statistician (1.4.4),

$$\mathrm{E}\{\mathrm{E}\{Y \mid X\}\} \;=\; \sum_x \mathrm{E}\{Y \mid X = x\}\, p_X(x).$$

(2) Substitute the conditional expectation (1.4.15), $\mathrm{E}\{Y \mid X = x\} = \sum_y y\, p_{Y|X}(y \mid x)$:

$$\mathrm{E}\{\mathrm{E}\{Y \mid X\}\} \;=\; \sum_x \Big(\sum_y y\, p_{Y|X}(y \mid x)\Big)\, p_X(x).$$

(3) Each outer term carries the factor $p_X(x)$, so the terms with $p_X(x) = 0$ vanish; in the terms with $p_X(x) > 0$, substitute the conditional pmf (1.4.10), $p_{Y|X}(y \mid x) = p(x, y)/p_X(x)$:

$$\mathrm{E}\{\mathrm{E}\{Y \mid X\}\} \;=\; \sum_x \Big(\sum_y y\, \frac{p(x, y)}{p_X(x)}\Big)\, p_X(x).$$

(4) The factor $p_X(x)$ does not involve y, so it passes inside the inner sum, where it cancels the denominator term by term,

$$y\, \frac{p(x, y)}{p_X(x)} \cdot p_X(x) \;=\; y\, p(x, y), \qquad \text{leaving} \qquad \mathrm{E}\{\mathrm{E}\{Y \mid X\}\} \;=\; \sum_x \sum_y y\, p(x, y).$$

(5) Exchange the order of summation (legitimate by the assumed absolute convergence), and pull y—constant in x—out of the inner sum:

$$\sum_x \sum_y y\, p(x, y) \;=\; \sum_y \sum_x y\, p(x, y) \;=\; \sum_y y \sum_x p(x, y).$$

(6) The inner sum is the marginal pmf of Y (1.4.8, with the roles of X and Y exchanged): $\sum_x p(x,y) = p_Y(y)$. Hence, by the definition of expectation (1.4.1),

$$\mathrm{E}\{\mathrm{E}\{Y \mid X\}\} \;=\; \sum_y y\, p_Y(y) \;=\; \mathrm{E}\{Y\}.$$

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Proof 1.4.16.b

Proof by Conditioning of the Iterated-Expectations Identity (continuous case).

(1) As before, $\mathrm{E}\{Y \mid X\}$ is a function of X, so by the law of the unconscious statistician (1.4.4), now in its integral form,

$$\mathrm{E}\{\mathrm{E}\{Y \mid X\}\} \;=\; \int \mathrm{E}\{Y \mid X = x\}\, f_X(x)\, dx.$$

(2) Substitute the conditional expectation (1.4.15), $\mathrm{E}\{Y \mid X = x\} = \int y\, f_{Y|X}(y \mid x)\, dy$:

$$\mathrm{E}\{\mathrm{E}\{Y \mid X\}\} \;=\; \int \left(\int y\, f_{Y|X}(y \mid x)\, dy \right) f_X(x)\, dx.$$

(3) The outer integrand carries the factor $f_X(x)$, so the set where $f_X(x) = 0$ contributes zero to the outer integral; on the set where $f_X(x) > 0$, substitute the conditional density (1.4.10), $f_{Y|X}(y \mid x) = f(x,y)/f_X(x)$:

$$\mathrm{E}\{\mathrm{E}\{Y \mid X\}\} \;=\; \int \left(\int y\, \frac{f(x,y)}{f_X(x)}\, dy \right) f_X(x)\, dx.$$

(4) The factor $f_X(x)$ does not involve y, so it passes inside the inner integral, where it cancels the denominator pointwise,

$$y\, \frac{f(x,y)}{f_X(x)} \cdot f_X(x) \;=\; y\, f(x,y), \qquad \text{leaving} \qquad \mathrm{E}\{\mathrm{E}\{Y \mid X\}\} \;=\; \iint y\, f(x,y)\, dy\, dx.$$

(5) Exchange the order of integration (legitimate, by Fubini's theorem, under the assumed absolute convergence), and pull y—constant in x—out of the inner integral:

$$\iint y\, f(x,y)\, dy\, dx \;=\; \iint y\, f(x,y)\, dx\, dy \;=\; \int y \left(\int f(x,y)\, dx \right) dy.$$

(6) The inner integral is the marginal density of Y (1.4.8, with the roles of X and Y exchanged): $\int f(x,y)\, dx = f_Y(y)$. Hence, by the definition of expectation (1.4.1),

$$\mathrm{E}\{\mathrm{E}\{Y \mid X\}\} \;=\; \int y\, f_Y(y)\, dy \;=\; \mathrm{E}\{Y\}.$$

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1.5 Decision Theory: Actions, Loss, and Risk

Section 1.3 left the clinical trial in a precise but incomplete state: a model (two independent Bernoulli samples, $\theta = (\theta_T, \theta_P) \in [0,1]^2$) and an analytic goal (learn the difference $\theta_T - \theta_P$). What it did not supply is a *method*—and candidate methods are cheap. Report the difference of the observed rates; report the shaded difference of 1.8.3; ignore the data and always report 0; average the first two. Every one of them produces an answer on every data set, so producing answers cannot be what "working" means. The deck's question—"But first, what do we mean 'good?' "—is not rhetorical: until *good* is defined, *best* is meaningless.

Decision theory is the discipline of making "good" explicit, and its move is to look past the answer to its *consequences*. Choosing is acting: report a number, declare efficacy, approve or shelve. Actions have consequences that depend on the unknown state of reality—report a difference of 0.30 when the truth is 0.05 and patients are steered to a nearly useless treatment; declare "no effect" when there is one and a working therapy is shelved. So attach to every (action, truth) pair a price: the *loss*. Judging a *method*—a rule, fixed before the data arrive, committing you to an action for every data set you might see—then means asking what it can be *expected* to lose. Because the data are random, that expected loss is computable from the model; because the truth is unknown, it must be computed at every candidate θ. The result is not a number but a curve over the parameter space: the *risk*.

A curve is an honest but awkward report card. Risk curves cross—the rule "always report 0" is unbeatable when $\theta_T = \theta_P$ and dreadful elsewhere—so "minimize the risk" cannot be a theorem-backed instruction (a principle, the deck insists, not a result). What survives is a discipline: discard any method whose curve another beats everywhere (admissibility), and, when a single number is wanted, average the curve against a prior (the Bayes risk of § 1.6). This framing of statistics—estimation and testing as one decision problem with different action spaces and losses (§ 1.8)—is Abraham Wald's, worked out in the 1940s, and the course adopts it from the outset. The definitions below fix the vocabulary; each is already visible in the trial. Decision theory begins from the statistical model of 1.3.1—a "parameterized set of distributions on the sample space"—and adds three objects. HPS 1

Definition 1.5.1

Action space

The *action space* $\mathcal{A}$ is the set of possible choices of the decision maker: the values an analysis is allowed to output.

Remark 1.5.2

Choosing the action space

The action space is fixed by the *question*, not by the data. The same trial supports estimation—report a plausible value of the cure-rate difference, $\mathcal{A} = [-1, 1]$, every value a difference of two proportions could take—or testing—assert "efficacious" or "not," an action space with exactly two elements; changing $\mathcal{A}$ changes the problem (§ 1.8 makes both versions precise). And note what $\mathcal{A}$ is not: it is not the sample space—data live in S, conclusions live in $\mathcal{A}$, and the method of 1.5.5 will be the bridge—and its elements carry no probabilities, because actions are *chosen*, not drawn.

Definition 1.5.3

Loss function

The *loss function* is a map

$$L : \mathcal{A} \times \Theta \ \to \ \mathbb{R},$$

where $L(a, \theta)$ tells you how much you lose by choosing action a when the parameter is θ.[4]

Remark 1.5.4

Reading the loss function

The loss is a *modeling choice*, as much a part of the problem's specification as the statistical model itself: it declares, in advance and in numbers, how bad each mistake would be—zero (or least) for the best action available at θ, growing as the action worsens, in whatever units of regret the analyst adopts. The course will stick to two standards (§ 1.8): *squared-error loss*, $L(a, \theta) = (a - \theta)^2$, for estimation—near-misses almost free, large misses punished disproportionately (double the miss, quadruple the loss)—and *zero–one loss* for testing, which charges 1 for a wrong assertion and 0 for a right one. Nothing in the definition demands symmetry—in the trial, shelving a working therapy may well cost more than approving a useless one, and a loss function can say so—but the two standards keep the theory clean. For the trial, with the difference as the target: $L(a, \theta) = \big(a - (\theta_T - \theta_P)\big)^2$.

Notice, too, where the analytic goal of 1.3.3 has re-entered the formalism. The model p^θ knows nothing of the study's purpose—it describes the data, whatever the study is for. The purpose lives entirely in the pair $(\mathcal{A}, L)$: the action space says what kind of answer the question demands, and the loss prices actions against θ only through the aspect of θ the goal names. The trial's loss just displayed touches $\theta = (\theta_T, \theta_P)$ only through the difference $\theta_T - \theta_P$—two truths with the same difference are indistinguishable to it—and in both of the course's special cases (§ 1.8) the same pattern holds: estimation's loss sees θ only through the estimand, testing's only through which side of the partition θ lies on. Change the goal—estimate θ_T alone, or merely test $\theta_T > \theta_P$—and the model stays put while $(\mathcal{A}, L)$ changes underneath it. That is how Figure 1.2.4's arrow from the analytic goal into the method is realized: the goal reaches the method through the loss it induces.

Definition 1.5.5

Method (decision rule)

A *method*, or *decision rule*, is a function

$$\delta : S \ \to \ \mathcal{A}$$

from the sample space to the action space: what you will do, specified in advance, for every data set you might observe.

[4] The deck words the domain both ways—"the parameter space cross the action space" on the definition slides, "the action space cross the parameter space" in the typeset cheat sheet—while always writing the arguments $L(a, \theta)$. These notes follow the cheat sheet.

Remark 1.5.6

A plan, not an answer

The force of the definition is in its timing. A method is chosen *before* the data arrive, and it commits you on every data set you might see—a complete contingency plan, not an answer. That is exactly what makes methods judgeable: a single answer cannot be evaluated until the truth is known, but a plan can be averaged over everything the model says the data may do. And since the data are modeled as random, $\delta(Y)$ is a random variable—so it has an expected loss, and that is where the risk comes from.

Example 1.5.7

Three methods for the trial

Each of the following is a method from $S = \{0,1\}^{n_T} \times \{0,1\}^{n_P}$ to $\mathcal{A} = [-1,1]$:

$$\delta_1(y) = \frac{x_T}{n_T} - \frac{x_P}{n_P}, \qquad \delta_2(y) = \frac{1+x_T}{2+n_T} - \frac{1+x_P}{2+n_P}, \qquad \delta_3(y) \equiv 0,$$

the difference of observed rates, the shaded version of 1.8.3, and a rule that ignores the data entirely. Nothing in the definition requires a method to be sensible—that is the point: "method" is the arena, and the rest of the section builds the referee.

Example 1.5.8

The same four objects, three more times

Nothing in 1.5.1–1.5.5 is about medicine. To see the skeleton *as* a skeleton, run three unrelated decisions through it.

(a) *The umbrella.* Unknown state: rain or not, $\Theta = \{\text{rain}, \text{dry}\}$. Data: the morning forecast probability, $S = [0,1]$. Actions: $\mathcal{A} = \{\text{carry}, \text{leave}\}$. The *entire* loss function is four numbers, visible at once:

$L(a,\theta)$	rain	dry
carry	1	1
leave	8	0

(carrying is the same mild nuisance either way; leaving is free when dry and a soaking when it rains). Decision rules are cutoffs: $\delta_c(y) = \text{carry}$ iff $y \geq c$—one rule per threshold c, a family that will reappear as the tests of § 1.8.

(b) *The spam filter.* Unknown state: whether the message is spam, $\Theta = \{\text{spam}, \text{legitimate}\}$. Data: the message itself ($S =$ all messages the system could receive). Actions: $\mathcal{A} = \{\text{quarantine}, \text{deliver}\}$. The loss is the umbrella's table with asymmetric stakes: quarantining a legitimate message costs, say, 10 (a missed offer, an angry colleague); delivering a spam message costs 1 (an annoyance); correct actions cost 0. The asymmetry is the modeling content—misprice the two mistakes and the "best" filter changes. Rules are again cutoffs: $\delta_t(\text{message}) = \text{quarantine}$ iff its score exceeds t.

(c) *The croissant order.* Unknown state: tomorrow's demand, $\Theta = \{0,1,2,\dots\}$. Data: past days' sales counts. Actions: the order quantity, $\mathcal{A} = \{0,1,2,\dots\}$—estimation-flavored, an action *is* a number. Loss, piecewise and asymmetric: $L(a,\theta) = c_{\text{o}}\,(a-\theta)^+ + c_{\text{u}}\,(\theta - a)^+$, where each unsold croissant costs its ingredients c_{o} and each missed customer costs the margin c_{u}. Candidate rules: order the historical average; order the average plus a safety cushion; always order forty.

The same quartet every time—an unknown state, the allowed outputs, priced consequences, and a data-to-action plan—and note what varied: $\mathcal{A}$ was two actions, two actions, then a count; L was a symmetric table, an asymmetric table, then a formula. The framework does not care, and the theory that follows applies to all of these at once.

Definition 1.5.9

Risk function

The *risk function* of a decision rule δ is the expected loss as a function of the parameter:

$$R_\delta(\theta) \;=\; \mathrm{E}^\theta\{L(\delta(Y),\theta)\} \;=\; \int_{y\in S} f^\theta(y)\,L(\delta(y),\theta)\,dy \quad \text{or} \quad \sum_{y\in S} p^\theta(y)\,L(\delta(y),\theta),$$

where E^θ denotes expectation when the data's distribution is the one indexed by θ.

Remark 1.5.10

Reading the risk function

Read the construction from the inside out: fix a candidate truth θ; the model hands the data their distribution; the method turns the random data into a random action $\delta(Y)$; the loss turns the random action into a random number $L(\delta(Y),\theta)$; and the expectation—the display in 1.5.9 is LOTUS (1.4.4) with $g(y) = L(\delta(y),\theta)$—collapses the randomness to a single number: the method's average penalty *if θ were the truth*. Sweeping θ across Θ traces the risk curve (Figure 1.5.11 shows the machine; Figure 1.5.14 shows the curves). Two instances carry the course: under zero–one loss the risk *is* the probability of a wrong assertion, $R_\delta(\theta) = P^\theta(\delta \text{ errs})$; under squared-error loss it is the mean squared error, which splits as variance plus squared bias (1.A.1). The trial's two estimators have their risks computed in closed form in 1.A.3 and 1.A.4.

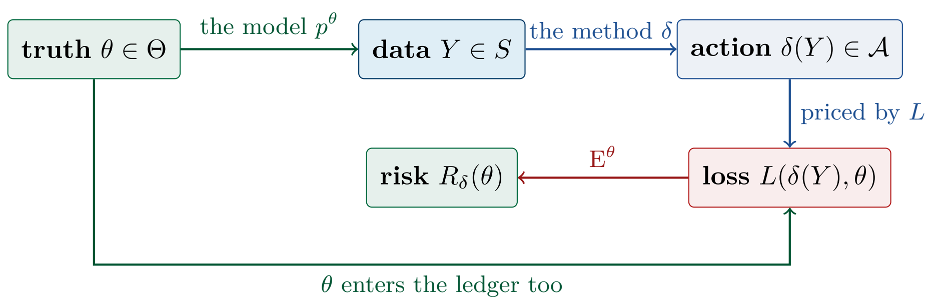

Figure 1.5.11. The anatomy of a decision problem. How the four objects compose, at one fixed candidate truth θ: the model hands the data their distribution, the method maps data to an action, the loss prices that action against θ, and the expectation over the data collapses the randomness to the single number $R_\delta(\theta)$. Sweeping θ over Θ traces the risk curves of Figure 1.5.14 (1.5.1, 1.5.3, 1.5.5, 1.5.9).

Principle 1.5.12

The risk principle

"THIS IS A PRINCIPLE, NOT A RESULT: try to minimize the risk." And—"Remember, Risk is a function!" One method's risk is an entire curve over Θ, not a number, so two methods need not be comparable: each may be better somewhere. The principle disciplines the search for methods; it does not by itself select one.

Definition 1.5.13

Admissibility

A method is *admissible* if there is no other method that does at least as well for every parameter value *and* better for some parameter value. A method that is so dominated is *inadmissible*: whatever θ turns out to be, you would not regret abandoning it.

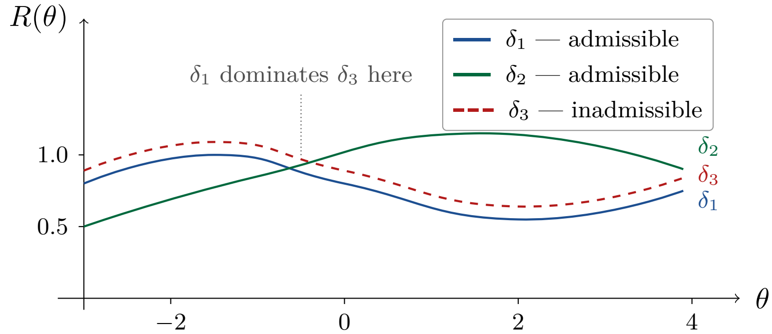

Figure 1.5.14. THREE RISK FUNCTIONS. Risk curves for three decision rules, as sketched in lecture. The dashed rule δ_3 lies strictly above δ_1 everywhere, so δ_1 dominates it and δ_3 is inadmissible; δ_1 and δ_2 cross, so neither dominates the other, and the deck's legend marks both admissible (1.5.13). A caution when comparing with the deck: its right-edge labels stack δ_1 above δ_3, pairing the names with the wrong endpoints; its legend and its own "δ_1 dominates δ_3" annotation—which require the blue curve to be the lower one—are authoritative, and are followed here (edge labels corrected to match the endpoints).

Concept 1.5.15

Should we follow the normative prescription?

If you have a coherent ordering on decision rules—any consistent way of ranking them—then that ordering is consistent with the risk for *some* loss function. "And so why not use THE loss function?!" Ranking methods at all commits you, implicitly, to behaving as if you were minimizing some risk; decision theory just makes the loss explicit.[5]

1.6 Bayesian Decision Theory

Section 1.5 ended with a deliberately incomplete verdict system. Admissibility only *prunes*: it discards the dominated methods and then falls silent, leaving a family of survivors whose risk curves cross—one better here, another better there (Figure 1.5.14). To pick a single method,

[5]Stated at the deck's informal level. A precise version—axioms of coherence and the accompanying complete-class theorems—is beyond this course.

something must say which regions of Θ matter most—how heavily the middle of the parameter space should weigh against its edges. Bayesian decision theory makes that weighting explicit, and makes it honest, by insisting it take the one lawful form a weighting of possibilities can take: a probability distribution on the parameter space. Everything else—the statistical model, the action space, the loss—stays exactly as § 1.5 built it; one object is added. HPS 1

Definition 1.6.1

Prior distribution

The *prior distribution* π is a probability distribution on the parameter space Θ: the parameter is treated as random, with π describing where it is likely to be before the data are seen.

Treating θ as random changes what the risk *is.* In § 1.5 the risk $R_\delta(\theta)$ was a curve—one number per candidate truth. With θ drawn from π, the curve's height becomes a random quantity, and a random quantity has an expectation: the collapse that turned random losses into the risk (1.5.10) can be run once more, one level up, averaging the curve itself. For the trial, the deck's choice is $\mathrm{Beta}(1, 1)$—the uniform distribution—on each arm's cure probability, the two priors independent (1.8.3).

Definition 1.6.2

Bayes risk

The *Bayes risk* of a decision rule δ under the prior π is the expected risk—"treating the parameter as random":

$$\mathrm{E}\{R_\delta(\theta)\} \;=\; \int_{\theta\in\Theta} \pi(\theta)\, R_\delta(\theta)\, d\theta$$

(a sum, for a discrete prior).[6] Where the risk was a function on Θ, the Bayes risk is a single number: the prior averages the curve.

A number, unlike a curve, orders the candidates completely: any two Bayes risks can be compared, and "best" finally means something. The prescription writes itself—and for the trial the arithmetic is carried out in 1.A.5, where, under the uniform priors, the shaded estimator of 1.8.3 turns out not merely to be good but to *be* the Bayes rule.

Principle 1.6.3

The Bayes principle

The Bayes version of the principle: minimize the *expected* risk. Among all decision rules, choose one whose Bayes risk is smallest—a *Bayes rule* for the prior π.

The principle is only as defensible as the distribution it consumes. Everything now turns on a question the mathematics cannot settle by itself: where does a prior come from?

Concept 1.6.4

Where does the prior distribution come from?

[6] The deck prints this integral without the trailing $d\theta$; it is restored here.

Maybe the parameter really is random and you know its distribution—or at least have some data to make a good guess about it. Or maybe you just make the prior up: "it can be helpful!"

Three provenances, running from the empirical down to the frankly invented—which raises the natural worry: does a prescription fed by an invented π deserve obedience? The deck closes the section with two defenses, one internal to the framework and one structural, tying the Bayes rules back to admissibility.

Remark 1.6.5

Should we follow the Bayes prescription?

If we believe in the prior, then there really is only one distribution for the data, and if we believe the general prescription (minimize risk), we should minimize the corresponding risk—the Bayes risk. Also: "(almost) every admissible rule is a Bayes rule and every Bayes rule is admissible... so even if we don't have a prior, the Bayes rule is sensible."[7]

1.7 The Cheat Sheet: Decision Theory in Eleven Items

Concept 1.7.1

The essential points

The instructor's own typeset summary of §§ 1.3–1.6, reproduced with the notation of these notes:

1. Sample space S.
2. Action space $\mathcal{A}$: the possible actions available to the decision-maker.
3. Parameter space Θ.
4. Data Y taking values in S.
5. There is a statistical model: for each θ, a distribution for Y, with f^θ or p^θ the corresponding density or pmf.
6. Loss function $L : \mathcal{A} \times \Theta \to \mathbb{R}$: $L(a, \theta)$ is the loss associated with making decision a when the true distribution is the one with pmf or density p^θ or f^θ.
7. A decision rule δ maps the sample space to the action space.
8. The risk function of δ, $R_\delta(\theta)$, maps each $\theta \in \Theta$ to the expected loss when the decision rule is δ and the distribution of Y is the one corresponding to θ:
$$R_\delta(\theta) = \mathrm{E}^\theta\{L(\delta(Y), \theta)\} = \int_{y \in S} f^\theta(y)\, L(\delta(y), \theta)\, dy \quad \text{or} \quad \sum_{y \in S} p^\theta(y)\, L(\delta(y), \theta).$$
9. Normative principle of decision theory: use only admissible decision rules—there must be no other rule at least as good for every θ and better for some θ.

[7] As stated in lecture, with the instructor's "(almost)." The two directions require side conditions in general; the precise statements are the complete-class theorems, beyond this course.

10. Prior distribution π for a random θ taking values in Θ; the Bayes risk of δ under π is

$$\mathrm{E}\{R_\delta(\theta)\} = \int_{\theta \in \Theta} \pi(\theta)\, R_\delta(\theta)\, d\theta.$$

11. The Bayesian principle: choose δ to minimize the Bayes risk.

1.8 Estimation, Testing, and the Clinical Trial

Two special cases of the framework carry the whole course. HPS 1

Definition 1.8.1

Estimation

Estimation is the special case in which the action space is the parameter space (or the set of values of the quantity being estimated[8]): the decision *is* a guess at the parameter. The loss function reflects a distance between the parameter and the estimate; this course will stick to *squared-error loss*,

$$L(a,\theta) \;=\; (a-\theta)^2.$$

Definition 1.8.2

Testing

Testing is the special case in which the action space has two choices: assert that the parameter is in one portion of the parameter space, or that it is in the rest. The loss function reflects whether you guessed right; this course will stick to *zero–one loss*,

$$L(a,\theta) \;=\; \mathbf{1}\{a \text{ is the wrong assertion about } \theta\},$$

which loses 1 for a wrong call and 0 for a right one.

Example 1.8.3

Two Bernoulli samples, and two estimators

The running example: two samples of independent Bernoulli outcomes with common expectations within each sample—θ_T in one arm, θ_P in the other. Estimate the difference $\theta_T - \theta_P$ between the two expectations, with squared-error loss. The Bayes version puts a $\mathrm{Beta}(1,1)$ (uniform) prior on each parameter, the two assumed independent. Two estimators will be compared:

1. the difference between the two observed rates, $X_T/n_T - X_P/n_P$; and
2. the difference if, separately in each sample, one instead uses

$$\frac{1+X}{2+n},$$

where X is the sum of the Bernoulli variables and n is the sample size.

[8] The deck says exactly "the action space is the parameter space"; the parenthetical broadening is ours, and is needed as soon as one estimates a *function* of the parameter—in the trial of 1.8.4, $\mathcal{A} = [-1,1]$ while $\Theta = [0,1]^2$.

Here the second estimator shades the observed rate toward $\frac{1}{2}$; where it comes from, and when it wins, is the business of the appendix (§1.A.4, §1.A.5).

Example 1.8.4

The clinical trial, formalized

For the two-arm trial of 1.3.4, with n_T treatment patients and n_P placebo patients and cure (1) or failure (0) recorded for each:

- *Sample space:* $S = \{0,1\}^{n_T} \times \{0,1\}^{n_P}$, the possible cure/failure records.
- *Statistical model and parameter space:* within each arm the outcomes are independent Bernoulli variables with a common expectation; the parameter is $\theta = (\theta_T, \theta_P)$ with $\Theta = [0,1]^2$, and

$$p^\theta(y) = \theta_T^{x_T}(1-\theta_T)^{n_T - x_T}\,\theta_P^{x_P}(1-\theta_P)^{n_P - x_P},$$

  where x_T, x_P are the arm totals of y.
- *Analytic goal:* learn the difference $\theta_T - \theta_P$ (is the treatment efficacious?).
- *Action space:* the possible values of the difference, $\mathcal{A} = [-1,1]$.
- *Loss function:* squared error, $L(a,\theta) = \big(a - (\theta_T - \theta_P)\big)^2$.

The lecture closes by naming, for each of the two estimators of 1.8.3: the risk function, the prior, and the Bayes risk. The deck sets these up and leaves the computations to us; they are worked in the appendix (§1.A.3–§1.A.5).

Remark 1.8.5

The probability theory we will need

The deck lists “Bias–variance decomposition” and “Beta distributions,” adding: “Also, variances, covariance, rules for expectations and variances for linear combinations of random variables; And the role of independence. . . ” For the Beta(α,β) distribution $(\alpha,\beta > 0)$ the deck displays the standard reference facts:[9]

$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{\mathrm{B}(\alpha,\beta)} \quad \text{on } [0,1], \qquad \mathrm{E}\{X\} = \frac{\alpha}{\alpha+\beta}, \qquad \mathrm{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)},$$

where $\mathrm{B}(\alpha,\beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$. Note that Beta$(1,1)$ is the uniform distribution on $[0,1]$.

Remark 1.8.6

What about AI?

The instructor’s advice: set up the calculations yourself—“that is practicing the conceptual issues, and that is the point”—while acknowledging that hand fluidity with such calculations “is already pretty much obsolete.” Two questions are left open: do you need to do some by hand to understand the conceptual side? And do you need to learn, now, how to get AI to do them for you?

[9] The deck shows the pdf/mean/variance rows of a standard reference table; they are typeset here directly.

Appendix

Supplementary material—additional proofs, context, and other treatments; not part of the lecture proper.

The deck names the bias–variance decomposition, the risk functions of the two estimators, and their Bayes risks (Slides 40–41 and the closing checklist), but works none of them. They are worked here, beyond the deck, in the deck's own notation. Throughout, $X \sim \text{Binomial}(n, \theta)$ denotes an arm total, so $\mathrm{E}\{X\} = n\theta$ and $\mathrm{Var}(X) = n\theta(1-\theta)$.

Before the first statement, its cast. The estimator $\delta = \delta(Y)$ is *random*—it inherits the data's randomness, so at each fixed candidate truth θ it has a whole sampling distribution of values it might take. The target $g(\theta)$ is, at that same fixed θ, just a *number* (in the trial, the difference $\theta_T - \theta_P$). Between the random estimator and the fixed target stand two more numbers, both functions of θ: $m(\theta) = \mathrm{E}^\theta\{\delta\}$, the estimator's own long-run average—where its sampling distribution centers—and $b(\theta) = m(\theta) - g(\theta)$, the *bias*: how far that center sits from the target. An estimator can therefore fail in exactly two ways: by scattering widely around wherever it centers (variance), and by centering in the wrong place (bias). The decomposition says that under squared-error loss, the risk is precisely these two failures added—no more, no less, and no interaction between them.

Supplement 1.A.1

The bias–variance decomposition

Let $\delta = \delta(Y)$ estimate a target $g(\theta)$ under squared-error loss, with $\mathrm{E}^\theta\{\delta^2\} < \infty$, and write $m(\theta) = \mathrm{E}^\theta\{\delta\}$ and $b(\theta) = m(\theta) - g(\theta)$ (the *bias*). Then

$$R_\delta(\theta) \;=\; \mathrm{E}^\theta\{(\delta - g(\theta))^2\} \;=\; \mathrm{Var}^\theta(\delta) \;+\; b(\theta)^2.$$

Proof 1.A.1

Proof of the Bias–Variance Decomposition by Adding and Subtracting the Mean.

(1) Route the miss through the estimator's own center: since $\delta - g = (\delta - m) + (m - g)$, every miss is a random scatter about the center plus the fixed offset of the center from the target.

(2) Square the two-term sum:

$$(\delta - g)^2 \;=\; (\delta - m)^2 \;+\; 2(\delta - m)(m - g) \;+\; (m - g)^2.$$

(3) Take E^θ term by term (linearity). The middle term vanishes: $m - g$ is a constant, so $\mathrm{E}^\theta\{2(\delta - m)(m - g)\} = 2(m - g)\,\mathrm{E}^\theta\{\delta - m\} = 0$, because deviations about one's own mean average to zero.

(4) What remains is $E^\theta\{(\delta - m)^2\} + (m - g)^2 = \mathrm{Var}^\theta(\delta) + b(\theta)^2$: the scatter term is the variance by definition, and the offset term is the squared bias.

■

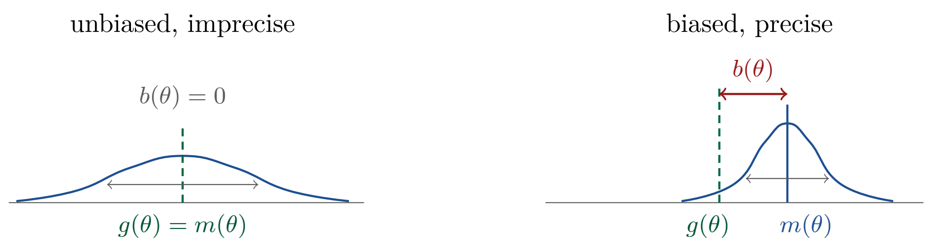

Figure 1.A.2. THE TWO WAYS AN ESTIMATOR CAN MISS. Two sampling distributions of an estimator δ against the same target $g(\theta)$ (dashed green). Left: centered on the target ($b(\theta) = 0$) but widely scattered—all its risk is variance. Right: tightly concentrated but systematically offset—its risk is a small variance plus the squared bias $b(\theta)^2$. Squared-error risk charges both sins at once, $R = \mathrm{Var} + b^2$ (1.A.1); neither picture dominates in general, and for the trial's two estimators which one wins depends on θ itself (1.A.3, 1.A.4; cf. the crossing curves of Figure 1.5.14).

The figure is the course's central trade in miniature, and the trial makes it numerical. The observed rate X/n is the left panel: unbiased, all of its risk variance, $\theta(1-\theta)/n$ (1.A.3). The shaded estimator $(1 + X)/(2 + n)$ is the right panel: it accepts the bias $(1 - 2\theta)/(n + 2)$ to shrink the spread (1.A.4). With $n = 10$: at $\theta = \frac{1}{2}$ the shaded risk is 0.0174 against the rate's 0.0250—the purchase pays—while at $\theta = 0.9$ it is 0.0107 against 0.0090—the purchase costs. Buying less variance with a little bias wins in the middle of the parameter space and loses at its edges: that is exactly why the two risk curves cross, why neither estimator is inadmissible, and why the prior's verdict in 1.A.5 depends on where it puts its weight.

Supplement 1.A.3

Risk of the difference of observed rates

For $\hat\delta = X_T/n_T - X_P/n_P$ in the model of 1.8.4:

$$R_{\hat\delta}(\theta) = \frac{\theta_T(1-\theta_T)}{n_T} + \frac{\theta_P(1-\theta_P)}{n_P}.$$

Proof 1.A.3

Proof of the Rate-Difference Risk via Unbiasedness and Independence.

(1) Each arm's rate is unbiased: $E^\theta\{X_T/n_T\} = n_T\theta_T/n_T = \theta_T$ (linearity), and likewise for the placebo arm; so $b(\theta) = E^\theta\{\hat\delta\} - (\theta_T - \theta_P) = 0$.

(2) A single Bernoulli variable Y has $E\{Y^2\} = E\{Y\} = \theta$, so $\mathrm{Var}(Y) = \theta - \theta^2 = \theta(1-\theta)$; summing n independent copies and scaling, $\mathrm{Var}(X/n) = \theta(1-\theta)/n$ for each arm.

(3) The arms are independent, so the variances of the difference add: $\mathrm{Var}^{\theta}(\hat{\delta}) = \theta_T(1-\theta_T)/n_T + \theta_P(1-\theta_P)/n_P$.

(4) By the bias–variance decomposition (1.A.1) with zero bias, the risk is the variance. ■

Uses 1.A.1.

Supplement 1.A.4

Risk of the shaded estimator

Write $T = \dfrac{1+X}{2+n}$ for one arm with $X \sim \mathrm{Binomial}(n, \theta)$. Then

$$\mathrm{E}^{\theta}\{T\} = \frac{1+n\theta}{n+2}, \qquad b(\theta) = \frac{1-2\theta}{n+2}, \qquad \mathrm{Var}^{\theta}(T) = \frac{n\theta(1-\theta)}{(n+2)^2},$$

so the one-arm risk is

$$\mathrm{MSE}_n(\theta) \;=\; \frac{n\theta(1-\theta) + (1-2\theta)^2}{(n+2)^2},$$

and for $\tilde{\delta} = T_T - T_P$ (the difference of the shaded rates, arms independent),

$$R_{\tilde{\delta}}(\theta) \;=\; \frac{n_T\theta_T(1-\theta_T)}{(n_T+2)^2} + \frac{n_P\theta_P(1-\theta_P)}{(n_P+2)^2} + \left(\frac{1-2\theta_T}{n_T+2} - \frac{1-2\theta_P}{n_P+2}\right)^2.$$

At $\theta = \frac{1}{2}$ the one-arm risk is $\frac{n}{4(n+2)^2} < \frac{1}{4n}$: shading toward $\frac{1}{2}$ beats the raw rate in the middle of the parameter space, at the price of bias near the edges—neither estimator dominates the other.

Proof 1.A.4

Proof of the Shaded Estimator's Risk by Linear-Combination Rules.

(1) By linearity, $\mathrm{E}^{\theta}\{T\} = (1 + \mathrm{E}^{\theta}\{X\})/(n+2) = (1+n\theta)/(n+2)$, so

$$b(\theta) = \frac{1+n\theta}{n+2} - \theta = \frac{1+n\theta-(n+2)\theta}{n+2} = \frac{1-2\theta}{n+2}.$$

(2) Adding constants does not change variance, and scaling squares: $\mathrm{Var}^{\theta}(T) = \mathrm{Var}(X)/(n+2)^2 = n\theta(1-\theta)/(n+2)^2$ (step (2) of 1.A.3).

(3) The one-arm risk is variance plus squared bias (1.A.1).

(4) For the difference, the arms are independent, so variances add, while the bias of a difference is the difference of the biases; 1.A.1 assembles the displayed risk.

(5) At $\theta = \frac{1}{2}$: bias $= 0$ and the risk is $\frac{n/4}{(n+2)^2}$, smaller than the raw rate's $\frac{1}{4n}$ since $n^2 < (n+2)^2$. ■

Uses 1.A.1, 1.A.3.

Supplement 1.A.5

Bayes risks under the uniform prior, and where $(1+X)/(2+n)$ comes from

Under the Beta$(1,1)$ (uniform) prior on each arm's parameter, independently:

1. the one-arm Bayes risk of the observed rate X/n is

$$\int_0^1 \frac{\theta(1-\theta)}{n}\,d\theta = \frac{1}{6n}\,;$$

2. the one-arm Bayes risk of $T = (1+X)/(2+n)$ is $1/\big(6(n+2)\big)$ — strictly smaller;
3. for the difference, the Bayes risks add across arms:

$$\frac{1}{6n_T} + \frac{1}{6n_P} \text{ for the rates,} \quad \text{versus} \quad \frac{1}{6(n_T+2)} + \frac{1}{6(n_P+2)} \text{ for the shaded estimators;}$$

4. and T is no accident: under the uniform prior, the posterior distribution of θ given $X = x$ is Beta$(1+x,\,1+n-x)$, whose mean is exactly $(1+x)/(n+2)$ — the shaded estimator is the *Bayes rule* for squared-error loss. (And since the arms' posteriors are independent, $T_T - T_P$ is the Bayes rule for the difference.)

Proof 1.A.5

Proof of the Bayes Risks by Integration, and of the Bayes Rule by Identifying the Posterior.

(1) $\int_0^1 \theta(1-\theta)\,d\theta = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$, so the rate's risk $\theta(1-\theta)/n$ integrates to $1/(6n)$.

(2) For T: $\int_0^1 (1-2\theta)^2\,d\theta = \int_0^1 (1-4\theta+4\theta^2)\,d\theta = 1 - 2 + \frac{4}{3} = \frac{1}{3}$, so

$$\int_0^1 \mathrm{MSE}_n(\theta)\,d\theta = \frac{n\cdot\frac{1}{6}+\frac{1}{3}}{(n+2)^2} = \frac{n+2}{6(n+2)^2} = \frac{1}{6(n+2)}.$$

(3) For the difference, expand the risk of 1.A.4 and integrate under the independent uniform priors. The cross term of the squared bias is

$$-2\int_0^1\!\int_0^1 \frac{(1-2\theta_T)(1-2\theta_P)}{(n_T+2)(n_P+2)}\,d\theta_T\,d\theta_P = 0,$$

because $\int_0^1 (1-2\theta)\,d\theta = 0$; what survives is the sum of the two one-arm Bayes risks (for the rates, the biases are identically zero).

(4) For the posterior: with prior density 1 on $[0,1]$ and $P(X = x \mid \theta) = \binom{n}{x}\theta^x(1-\theta)^{n-x}$, the posterior density is proportional to $\theta^x(1-\theta)^{n-x} = \theta^{(1+x)-1}(1-\theta)^{(1+n-x)-1}$ — the Beta$(1+x, 1+n-x)$ kernel. Its mean, by the Beta facts of 1.8.5, is $(1+x)/(n+2)$.

(5) Finally, the action a minimizing the posterior expected loss $\mathrm{E}\{(\theta-a)^2 \mid X = x\} = \mathrm{Var}(\theta \mid X = x) + (\mathrm{E}\{\theta \mid X = x\} - a)^2$ (by 1.A.1 applied to the posterior) is the posterior mean; and since the Bayes risk is the expectation over X of the posterior expected loss (iterated expectations, 1.4.16), minimizing the latter for each x minimizes the former, so T is the Bayes rule.

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Uses 1.A.3, 1.A.4, 1.8.5.

Index of Objects